

JESUS MANCILLA

Senior Applied Scientist | AI Evaluation & GenAI Quality Systems

Evaluation-driven development: offline benchmarks, online monitoring, HITL scoring, and LLM-as-judge pipelines
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PROFESSIONAL SUMMARY

Senior applied scientist and platform builder focused on **evaluation systems for AI-driven experiences**. Builds offline benchmark suites, automated regression tests, and human-in-the-loop scoring loops (including **LLM-as-judge** and verifier-driven refinement) to support eval-driven development. Delivered reliability and performance gains, including **~33% hallucination reduction** and **p95 TTFT 4.3s to under 2s** (internal), and reduced open-ended analysis from **~30h to under 8h** and document workflows from **~90m to under 5m**. Strong cross-functional partner, known for reusable tooling, clear documentation, and adoption. Bilingual (English/Spanish).

TECHNICAL STACK & DOMAINS

AI Evaluation & Quality: Offline benchmarks, online monitoring, acceptance tests, curated benchmark datasets, label taxonomy refinement, human-in-the-loop review, LLM-as-judge, prompt and evaluation criteria iteration
Programming: Python, SQL, JavaScript, TypeScript, Git
LLM Systems: Agents (LangGraph), RAG, vector search, multi-agent orchestration, observability and monitoring
Data & Experimentation: Regression, ANOVA, t-tests, online experiments (A/B testing), longitudinal tracking, large-scale log analysis, KPI definition
Frameworks: LangChain, LangGraph, FastAPI, React, Next.js
Domains & Scale: Consumer media (500M+ MAU), TV/streaming (~70M devices), education, commerce, automotive/HF

WORK EXPERIENCE

AnyTickets	Houston, TX (Remote)
Lead AI Developer (prev. Senior AI Developer)	Mar 2026 – Present
- Audited production model inputs and found ~80% of comparables were the company’s own listings (circular pricing) and the assumed distribution empirically wrong, redirecting the algorithm. - Built a golden-set evaluation harness over a hand-verified 696-section corpus (30 venues), lifting automated seat-map coverage from ~5% to ~95%. - Ran a controlled A/B against the manual pricing baseline on split inventory: 34.8% vs 30.2% sell-through and 42.8% vs 34.5% price realization. - Established the team’s agentic-engineering practice (nested context architecture, multi-model adversarial review), now taught to engineering teams outside the org.	
Argomai	Houston, TX (Remote)
Senior Applied Scientist	Jan 2025 – Mar 2026
- Built and governed enterprise AI evaluation and deployment processes, defining review loops, monitoring expectations, and quality gates across client projects. - Authored and open-sourcing a multilingual QA evaluation pipeline (self-consistency → critique → refinement) with judge-centric scoring and deterministic audit checks. - Designed reusable platform components (embeddings, prompt templates, orchestration SDK) and agent workflows to standardize evaluation and reuse across initiatives. - Reduced document workflow cycle time from ~90m to under 5m and PM reporting from 6h/wk to under 1h via retrieval automation and structured aggregation.	
Meta	Houston, TX (Remote)
Senior Quantitative UX Researcher	Jan 2024 – Jan 2025
- Implemented evaluation pipeline patterns, including LLM-as-judge self-reflection and verifier-driven revisions, improving reliability of intermediate outputs for production research workflows. - Designed human-in-the-loop expert review to refine outputs in natural language and curate few-shot examples, reducing analysis time ~73% (~30h to under 8h). - Engineered multi-agent sampling (self-consistency with chain-of-thought) to generate and score candidate outputs across tasks, enabling hill-climbing style iteration. - Ran pre and post-launch monitoring using bi-weekly surveys and SQL-based log analysis to track model quality changes and guide roadmap decisions at 500M+ MAU scale.	

Roku	San Jose, CA
Senior User Experience Researcher	Jan 2021 – Nov 2023
– Built a modular survey analysis and reporting engine, cutting weekly report generation from ~4h to under 5m via automation and reusable analysis logic. – Developed an AI-powered indexed research database enabling self-serve discovery and faster executive reporting across orgs.	
Walmart Global Tech	Sunnyvale, CA
Senior User Experience Researcher	Aug 2019 – Nov 2020
– Owned product analytics for Sam's Club mobile, defining KPIs and dashboards that linked behavioral and business metrics to feature decisions. – Standardized cross-team research operations using Jira and Confluence workflows to prioritize requests, track WIP, and improve delivery predictability.	
Scrapworks Inc.	Palo Alto, CA
Data Scientist	Sep 2017 – Aug 2019
– Delivered deep learning forecasting for commodities, reducing prediction error by 60% and improving decision support for traders. – Initiated an NLP-based merchandise classifier and contributed to a patent application, building end-to-end data preparation and modeling workflows.	
Suggestic	Mexico City, Mexico
Senior User Experience Researcher	Dec 2016 – Sep 2017
– Ran data-driven testing and analysis for new features, using analytics to iterate quickly and improve engagement in a consumer coaching app. – Transitioned core product experience from conversational to graphical UI using insights from experimentation and behavioral signals.	
Stanford University	Stanford, CA
User Experience Researcher	May 2016 – Nov 2016
– Conducted ML research on stress detection with 150+ hours of multimodal data and contributed to models reaching ~90% accuracy.	
ITAM	Mexico City, Mexico
User Experience Researcher	Aug 2014 – May 2016
– Analyzed psychophysiological signals and built custom visualizations, applying ML to surface patterns and inform user-centered product decisions.	

SELECTED PROJECTS (AI EVALUATION & ML PLATFORMS)

- Evaluation Benchmarking Suite:** Built automated regression tests on a **500+ item** dataset to track adherence, tone, and reasoning; improved reliability (about **33%** hallucination reduction) and reduced p95 TTFT from **4.3s to under 2s** (internal).
- Quant UX at Scale:** 6-step GenAI evaluation pipeline (embeddings, self-consistency, LLM-as-judge, verifier revisions, HITL, few-shot curation), cutting time per study from **~30h to under 8h** in a **500M+ MAU** context.
- Modular Survey Analysis System:** Production reporting engine with automated stats and classifier prototyping, reducing weekly reporting from **~4h to under 5m**.

EDUCATION

Instituto Tecnológico Autónomo de México (ITAM)	
M.S. in Computer Science (HCI/AI Focus)	2014 – 2016
Universidad de Colima	
B.A. in Psychology	2009 – 2013

SELECTED PUBLICATIONS

Ramos-Rivera, R. E., Santana Mancilla, P. C., Garcia-Mancilla, J., & Gaytán-Lugo, L. S. (2025). Language models in education: Generative AI to optimize teacher performance analysis. *InnovAcademica*, 1(2), 74–85.

Ramos-Rivera, R. E., Garcia-Mancilla, J., Cárdenas-Villa, G. E., & Santana-Mancilla, P. C. (2024). Towards Improving Teacher Performance Assessment through Human-Centered AI-Powered Survey Analysis. *Avances en Interacción Humano-Computadora*, 9(1), 261–264.

Garcia-Mancilla, J., Ramirez-Marquez, J. E., Lipizzi, C., Vesonder, G. T., & Gonzalez, V. M. (2018). Characterizing negative sentiments in at-risk populations via crowd computing. *International Journal of Data Science and Analytics*.

For full list, see: jgmancilla.com/research-papers